

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (VDE) (CBCS) Degree Examinations

May-June 2025

22VDE202 – REAL TIME OPERATING SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five** full questions choosing **One** full question from each Unit.

Unit – I

	Marks	BT*	CO*	PO*
1. a) Discuss preemptive versus non-preemptive schedulers. Also provide their taxonomy.	10	L*2	1	3
b) Prove that RM is an optimal scheduling policy under static priority algorithm using examples.	10	L3	1	3
2. a) Discuss the service utility for the following real-time services. i) Soft real-time service.	10	L2	1	3
ii) Isochronal service.	10	L2	1	3
b) Explain the state transition table for a thread of execution with necessary diagrams.	10	L2	1	3

Unit – II

3. a) Explain Deadline Monotonic scheduling policy. Provide the sufficient feasibility test for the same with relevant equations.	10	L2	2	3
b) Given three services S1, S2, and S3 with service release periods T1=3, T2=5, and T3=16, and service execution times C1=1, C2=1, and C3=3. Draw schedule diagram to find if the service set is schedulable under EDF and LLF.	10	L3	2	3
4. a) How scheduling point test and completion time test are conducted to decide the feasibility of a schedule?	10	L2	2	3
b) For the given task set, illustrate the multiprocessor scheduling using First-fit decreasing method under Bin-packing assignment algorithm.				

	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11
e _i	5	7	3	1	10	16	1	3	9	17	21
P _i	10	21	22	24	30	40	50	55	70	90	95

10 L3 2 3

Unit – III

5. a) Discuss the hazards that may stall the pipeline and increase CPI of a real time system.	10	L2	3	3
b) Explain Set associative mapping of memory to cache. Compare the same with direct mapping of memory to cache.	10	L2	3	3
6. a) Draw block diagrams for physical memory hierarchy using Harvard architecture and its logical partitioning and segmenting by firmware. With neat sketches, explain briefly direct mapping of memory to caches.	10	L2	3	3
b) Given a data word 11000100, perform hamming encoding for the following cases. i) No errors in the received data.	10	L2	3	3
ii) Error in parity word.				

P.T.O

Unit – IV

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| 7. | a) | Discuss on Trace Ports. Explain in the concept of Heisenbug. | 10 | L2 | 4 | 3 |
| | b) | Provide a simple instructive example of the use of a binary semaphore by two service tasks in the VxWorks RTOS framework. | 10 | L3 | 4 | 3 |
| 8. | a) | Explain the following firmware components of an RTOS. | | | | |
| | i) | Device drivers. | 10 | L2 | 4 | 3 |
| | ii) | Operating System Services. | 10 | L3 | 4 | 3 |
| | b) | Define exceptions. Explain with simple example code how exceptions arise and the use of asserts in handling exceptions. | 10 | L3 | 4 | 3 |

Unit – V

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|-----|----|---|----|----|---|---|
| 9. | a) | With a neat diagram, explain the concept of Drill-Down Tuning. | 10 | L2 | 5 | 3 |
| | b) | Write a C-code to compute the Fibonacci sequence. | 10 | L2 | 5 | 3 |
| 10. | a) | Discuss the methods for building performance monitoring capability into the software. | 10 | L2 | 5 | 3 |
| | b) | Discuss the similarities and differences between Reliability and Availability. | 10 | L2 | 5 | 3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
